ECA56 – ADVANCED WIRELESS COMMUNICATION SYSTEMS

ASSIGNMENT – 1

CO1: Cellular Network Evolution, Cellular Planning, Spectrum Allocation

Item	Details
Course	ECA56 – Advanced Wireless Communication Systems
Assignment	Assignment – 1
Topic	Cellular Network Evolution, Cellular Planning, Spectrum Allocation
Assessment	Open-ended Engineering Design Report
Maximum Marks	100
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1. Executive Summary

This report develops a practical 5G/6G-ready wireless communication design for Tamil Nadu, Karnataka, Kerala, Maharashtra, Gujarat and Uttar Pradesh. The design supports enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), massive Machine-Type Communications (mMTC), autonomous transportation, remote healthcare, industrial automation, high-definition video, emergency services and massive IoT.

A heterogeneous network is selected because the six states have different population densities, terrain, industrial activity and traffic requirements. Sub-1 GHz is assigned mainly to wide-area rural, transport and public-safety coverage; mid-band spectrum is the principal layer for balanced coverage and capacity; and mmWave is used selectively in high-capacity hotspots, industrial campuses and dense urban locations.

The proposed architecture combines macro gNodeBs, small cells, 5G Core functions, edge computing, network slicing, massive MIMO, beamforming, OFDMA, AI-assisted radio-resource management and resilient backhaul. A weighted decision matrix gives the heterogeneous architecture a higher overall score than a macro-cell-focused alternative.

The assignment does not provide numerical subscriber count, busy-hour traffic per subscriber or available traffic-channel values. Therefore, the numerical examples in this report use clearly stated engineering assumptions. These values are illustrative and must be replaced by measured/operator planning data before deployment.

2. Problem Statement and Design Objectives

The consortium requires an integrated communication network across six states containing urban, suburban, rural, industrial and transport environments. The network must balance coverage, capacity, latency, reliability, spectrum efficiency, interference, energy use, security, public safety and deployment cost.

Objective	**Engineering target / design direction**
Coverage	Continuous basic coverage along populated areas, highways and critical public-safety corridors.
Capacity	Provide high-capacity mid-band and selective mmWave layers in traffic hotspots.
Latency	Use local edge processing for URLLC and latency-sensitive applications.
Reliability	Provide redundant transport, prioritized slices and resilient core services.
Spectrum efficiency	Use OFDMA, massive MIMO, beamforming, reuse and dynamic allocation.
Interference	Apply power control, beam management, coordinated scheduling and frequency planning.
Security	Use strong authentication, encryption, anomaly detection and recovery mechanisms.
Energy	Use sleep modes, adaptive power, efficient resource allocation and infrastructure sharing.
Scalability	Use cloud-native core, modular RAN and scalable edge resources.

3. Stage 1 – Requirements and Problem Formulation

3.1 State-wise Requirements Matrix

State	Major environments	Main communication demand	Main challenge	Initial design target
Tamil Nadu	Dense cities, manufacturing, highways, ports	eMBB, industrial automation, transport, IoT	High traffic and industrial reliability	Dense mid-band + small cells; mmWave hotspots; edge at industrial/urban sites
Karnataka	Urban technology centres, industrial areas, highways, rural regions	eMBB, cloud services, autonomous transport, IoT	High urban capacity and mobility	Mid-band macro/small cells + edge; sub-1 GHz rural layer
Kerala	Dense linear settlements, coastal and hilly regions	Broadband, healthcare, emergency services, IoT	Terrain, distributed settlements and disaster resilience	Sub-1 GHz coverage + mid-band capacity; resilient backhaul
Maharashtra	Very large urban/industrial regions, highways	High-density broadband, finance, industry, transport	Extreme capacity demand and congestion	Dense heterogeneous RAN; mid-band + selective mmWave + edge
Gujarat	Industrial corridors, cities, ports, rural areas	Industry, logistics, broadband, IoT	Industrial reliability and wide-area connectivity	Mid-band industrial layer + sub-1 GHz coverage + mmWave hotspots
Uttar Pradesh	Large rural and urban population, highways, cities	Broadband, public services, healthcare, IoT	Large geographic area and coverage cost	Sub-1 GHz macro coverage + mid-band capacity; targeted small cells

3.2 Application Classification

Application	Primary service class	Key requirement	Network response
HD video / mobile broadband	eMBB	High throughput	Mid-band, massive MIMO, small cells and selective mmWave
Autonomous transportation	URLLC	Very low latency, high reliability, mobility continuity	Edge computing, prioritized slice, beam management and redundant transport
Remote healthcare	URLLC/eMBB	Reliability, low latency and high-quality media	Dedicated slice, edge processing and resilient backhaul
Industrial automation	URLLC/mMTC	Reliability, deterministic response and many devices	Private/enterprise RAN where appropriate, edge and prioritized resources
Massive IoT	mMTC	Large device population and energy efficiency	Wide-area coverage, efficient scheduling and IoT-optimized resources

Emergency communication	URLLC/public safety	Priority, availability and resilience	Dedicated slice, traffic pre-emption and redundant network resources

3.3 Constraints and Conflicting Requirements

- Coverage versus capacity: low-frequency spectrum covers larger areas, while high-frequency spectrum provides more bandwidth but shorter practical range.
- Capacity versus cost: more cells increase capacity but require additional sites, power, backhaul and maintenance.
- Latency versus centralized processing: centralized clouds are efficient, but edge computing reduces transport and processing delay.
- Energy versus performance: increasing transmit power and cell density can improve service but increases energy consumption.
- Reuse versus interference: aggressive frequency reuse improves spectrum utilization but requires better interference coordination.
- Security versus operational complexity: stronger security controls improve resilience but add processing, monitoring and management requirements.

3.4 Constrained Optimization Formulation

The design can be expressed as a multi-objective optimization problem. Let x represent cell locations, spectrum assignments, transmit powers, antenna configurations, routing choices, slice resources and edge locations.

Minimize: $J(x) = w1 \cdot CoverageGap + w2 \cdot Congestion + w3 \cdot Latency + w4 \cdot Energy + w5 \cdot Cost + w6 \cdot Interference + w7 \cdot ReliabilityPenalty$.

Subject to: coverage ≥ target; blocking probability ≤ 0.02; latency ≤ application limit; reliability ≥ required value; available spectrum ≤ licensed allocation; transmit power ≤ regulatory limit; and deployment cost ≤ budget.

The weights are selected according to service priority. Public safety and URLLC receive higher reliability and latency priority, while eMBB emphasizes throughput and capacity.

3.5 Performance Indicators

KPI	**Purpose**
Coverage probability	Measures whether users and critical corridors receive usable service.
Throughput per user	Measures eMBB experience.
Cell utilization	Identifies congestion and underused capacity.
Blocking probability	Measures probability that offered traffic cannot be served.
Latency	Measures suitability for URLLC and edge applications.
Packet loss	Measures transport and application reliability.
Spectral efficiency	Measures bits/s/Hz performance.
Energy per delivered bit	Measures sustainability.
Handover failure rate	Measures mobility continuity.
Security incident detection time	Measures resilience and operational response.

3.6 Design Decision 1

The most critical requirements are coverage, capacity, reliability, latency and cost. The proposed priority is: (1) public-safety and critical-service coverage, (2) reliable low-latency services, (3) sufficient broadband capacity, (4) scalable IoT support, and (5) energy and cost optimization. No single requirement is optimized independently because increasing cell density or bandwidth can increase cost, energy use and interference-management complexity.

4. Stage 2 – Spectrum, Radio Access and Cellular Planning

4.1 Spectrum Comparison

Spectrum	Coverage	Capacity	Propagation / blockage	Recommended use
Sub-1 GHz	Excellent relative wide-area coverage and penetration	Lower bandwidth availability	Lower frequency generally propagates farther	Rural coverage, highways, IoT, emergency communication
Mid-band 1–6 GHz	Good balance	High and practical capacity	Moderate range and blockage	Main urban/suburban/industrial/transport layer
mmWave >24 GHz	Shorter coverage	Very high bandwidth potential	High blockage sensitivity and practical range limitations	Dense hotspots, industrial sites, fixed high-throughput links

A uniform spectrum band is not appropriate because the six states contain different traffic density and geographic conditions. A layered spectrum strategy provides coverage from sub-1 GHz, capacity from mid-band and extreme hotspot capacity from mmWave.

4.2 Cellular Architecture

Cell type	Typical role in proposed network
Macro cell	Wide-area coverage, mobility anchor, rural/suburban service
Micro/small cell	Urban streets, transport hubs and medium-density capacity enhancement
Pico/small cell	Buildings, campuses, enterprise areas and localized hotspots
mmWave small cell	Very high traffic zones and short-range high-throughput service
Edge site	Local processing for latency-sensitive applications

4.3 Radio Technologies

- OFDMA divides the available spectrum into orthogonal resource units and allows flexible allocation among users.
- Massive MIMO uses many antenna elements to improve spatial multiplexing, coverage and capacity.
- Beamforming directs radio energy toward intended users, improving link quality and reducing unnecessary interference.
- Heterogeneous networks combine macro and small cells so that coverage and capacity can be optimized independently.
- Frequency reuse allows the same spectrum to be reused in geographically separated cells. Modern systems can use aggressive reuse with interference coordination.

4.4 Capacity Estimate and Cell Count

Illustrative assumption for planning: an average busy-hour effective cell capacity of 500 Mbps is used for a mixed urban/suburban cell. If an area generates 4,000 Mbps of busy-hour demand, the theoretical minimum number of such cells is:

N = Total traffic demand / Effective cell capacity = 4000 / 500 = 8 cells.

A planning margin of 25% is then added for growth, traffic variation and operational headroom: Nplanned = ceil(8 × 1.25) = 10 cells.

This is a simplified capacity calculation. Actual cell counts must also satisfy radio coverage, propagation, terrain, backhaul and mobility constraints. Therefore, coverage-driven and capacity-driven cell counts should both be calculated and the larger requirement should guide deployment.

4.5 Cell Splitting and Sectoring

Cell splitting reduces the coverage area of a congested cell by deploying additional smaller cells. It increases spatial reuse and therefore capacity, but also increases site cost, backhaul requirements, interference coordination and energy consumption. Sectoring divides a cell using directional antennas. It can improve frequency reuse and reduce interference in the desired direction without requiring as many additional sites.

4.6 Interference Analysis and Mitigation

Interference	Cause	Mitigation
Co-channel	Same frequency reused in nearby cells	Beamforming, power control, coordinated scheduling, cell planning and reuse optimization
Adjacent-channel	Energy leakage between neighboring channels	Filtering, guard allocation where required and careful frequency planning
Inter-cell interference	Overlapping transmissions	ICIC/coordination, dynamic resource allocation and beam management
Small-cell interference	Dense heterogeneous deployment	Adaptive power, topology-aware scheduling and interference-aware association
Blockage / beam instability	mmWave obstacles and mobility	Beam tracking, multiple candidate beams and fallback to lower-frequency layer

4.7 State-wise Spectrum and RAN Planning

State	Sub-1 GHz	Mid-band	mmWave	RAN strategy
Tamil Nadu	Rural/highway	Primary layer	Industrial and dense urban hotspots	Macro + small cells + edge
Karnataka	Rural/transport	Primary urban layer	Technology/enterprise hotspots	Macro + small cells + edge
Kerala	Wide-area/disaster	Urban and corridor capacity	Selected dense locations	Coverage-first heterogeneous RAN
Maharashtra	Wide-area backup	Primary capacity layer	Major urban hotspots	Dense small-cell and edge deployment
Gujarat	Rural/industrial coverage	Industrial/city layer	Ports and high-throughput hotspots	Macro + industrial small cells + edge

Uttar Pradesh	Primary wide-area layer	Urban/corridor capacity	Selected hotspots only	Coverage-focused macro + targeted small cells

4.8 Design Decision 2

The recommended architecture is a multi-band heterogeneous RAN. Sub-1 GHz provides the coverage foundation; mid-band provides the main capacity layer; and mmWave is selectively deployed where traffic density justifies its shorter range and higher site density. This avoids the cost of deploying mmWave uniformly and avoids the capacity limitations of a sub-1 GHz-only design.

5. Stage 3 – Quantitative Capacity, Latency and Reliability Models

5.1 Illustrative Traffic Demand Model

Because the question paper does not provide numerical subscriber count or per-subscriber busy-hour traffic, the following values are explicitly assumed for demonstration.

Application	Number of active users/devices	Average busy-hour demand	Aggregate demand
HD video / eMBB	10,000 users	2 Mbps/user	20,000 Mbps = 20 Gbps
Autonomous transport	2,000 vehicles	1 Mbps/vehicle	2,000 Mbps = 2 Gbps
Industrial IoT	100,000 devices	0.02 Mbps/device	2,000 Mbps = 2 Gbps

Total illustrative busy-hour demand = 20 + 2 + 2 = 24 Gbps. A 25% engineering margin gives 30 Gbps planning capacity. This margin is not a universal standard; it is an explicit planning assumption.

5.2 Spectrum and Spectral Efficiency

Using $R = B\eta$, where R is achievable data rate, B is bandwidth and η is spectral efficiency, a 30 Gbps aggregate target could be supported through multiple cells and bands rather than a single channel.

For example, if an effective average spectral efficiency of 5 bit/s/Hz is assumed, the ideal aggregate bandwidth equivalent is B = 30 Gbps / 5 bit/s/Hz = 6 GHz. This does not mean 6 GHz of contiguous spectrum is required at one location; the network obtains capacity through spatial reuse, multiple carriers, multiple cells and multiple bands.

The Shannon relationship $C = B \log_2(1 + SNR)$ gives a theoretical upper bound. Practical throughput is lower due to interference, implementation losses, protocol overhead and scheduling.

5.3 Propagation, Radius and User Distribution

Cell radius should be selected using both link-budget coverage and traffic density. A low-frequency macro cell can cover a larger area, while a mid-band or mmWave small cell generally covers a smaller area. Urban high-rise environments and industrial structures can create shadowing and blockage. Therefore, the network uses overlapping layers: a coverage layer and capacity layers.

If traffic density doubles while the available capacity per cell remains constant, cell utilization doubles. Once utilization approaches a congestion threshold, the planner can add small cells, increase available spectrum, improve spectral efficiency or offload users to another layer.

5.4 Latency Budget

Component	Illustrative budget	Engineering action
Radio access	5–10 ms	Use suitable scheduling, beam management and reliable radio links

Transport/backhaul	5–10 ms	Use efficient routing and local breakout
Core processing	2–5 ms	Cloud-native functions and optimized user plane
Edge application processing	2–5 ms	Place edge resources near traffic source
Illustrative end-to-end target	~20–30 ms for selected applications	Tune according to actual application requirement

Critical URLLC applications may require tighter targets than this illustrative budget. Actual latency depends on application, packet size, scheduling, processing and transport conditions. Edge computing is therefore essential for applications such as vehicle coordination and industrial control.

5.5 Mobility and Handover

Autonomous vehicles moving along highways require continuous service while crossing cells. The network should maintain overlapping coverage, use measurement-based handover, prepare target cells before handover and prioritize safety traffic. Mid-band can provide capacity while sub-1 GHz provides a robust coverage fallback. Edge sites should be distributed along major transport corridors to reduce the distance to latency-sensitive applications.

5.6 Erlang B Traffic-Channel Calculation

The Erlang B model estimates the probability that an offered traffic call/session is blocked when all available channels are busy. The blocking probability is:

$B(E,N) = (E^N / N!) / \Sigma(k=0 \text{ to } N) (E^k / k!)$, where E is offered traffic in Erlangs and N is the number of traffic channels.

The assignment requests a target blocking probability of 2%. Since subscriber count and busy-hour traffic per subscriber are not numerically supplied in the question paper, a reproducible illustrative example is used: 1,000 active subscribers, 0.02 Erlang busy-hour traffic per subscriber, giving E = 20 Erlangs.

For E = 20 Erlangs, the Erlang-B recursion $B_N = (E \cdot B_{(N-1)})/(N + E \cdot B_{(N-1)})$ gives the minimum illustrative channel count N = 28, with blocking probability ≈ 1.88%. Therefore, at least 28 traffic channels are required under the stated illustrative assumptions.

Channels N	**Blocking probability (approx.)**
20	15.89%
22	10.67%
24	6.61%
25	5.02%
26	3.72%
27	2.68%
28	1.88%

The result must be recalculated using the actual subscriber count, busy-hour traffic and channel definition supplied by the operator. Guard bands, duplexing overhead, control channels and other practical spectrum overheads are ignored in this idealized channel-count example, as required to keep the Erlang calculation focused on traffic-channel capacity.

5.7 Congestion Thresholds

Metric	**Suggested planning threshold**	**Action**
Average cell utilization	≤ 70%	Normal operation and monitoring
Busy-hour utilization	70–85%	Prepare optimization/offload
Sustained utilization	> 85%	Add capacity, spectrum or small cells

Packet loss for critical traffic	Very low / application-specific	Prioritize traffic and investigate transport/RAN
Handover failure	Low and continuously monitored	Optimize neighbor relations, coverage and mobility parameters

5.8 Design Decision 3 – Main Bottlenecks by State

State	Likely bottleneck	Selected intervention
Tamil Nadu	Urban/industrial capacity	Small cells, mid-band expansion, edge and interference coordination
Karnataka	Urban peak traffic and mobility	Mid-band densification, edge and transport-corridor optimization
Kerala	Coverage and resilience	Sub-1 GHz expansion, redundant backhaul and edge at critical locations
Maharashtra	Very high urban capacity	Dense small cells, additional mid-band capacity and selective mmWave
Gujarat	Industrial reliability and capacity	Industrial small cells, edge and resilient transport
Uttar Pradesh	Large-area coverage and cost	Sub-1 GHz macro layer, targeted mid-band and hotspot densification

6. Stage 4 – Alternative Network Architectures

6.1 Architecture A – Macro-Cell Focused

Architecture A uses mainly sub-1 GHz and mid-band macro cells. It provides broad coverage with fewer sites and relatively simpler operations. However, it has limitations in very dense urban traffic, industrial campuses and mmWave-class hotspot capacity.

6.2 Architecture B – Heterogeneous 5G/6G-Ready

Architecture B combines sub-1 GHz macro cells, mid-band macro/small cells, selective mmWave small cells, massive MIMO, beamforming, edge computing, network slicing and AI-assisted resource management. It has higher deployment complexity and cost but provides better scalability and service differentiation.

6.3 Weighted Decision Matrix

Criterion	Weight	A: Macro-focused	B: Heterogeneous
Coverage	20%	5	4
Capacity	20%	3	5
Latency	15%	3	5
Reliability	15%	4	5
Scalability	10%	3	5
Deployment simplicity	5%	5	3
Energy efficiency	5%	4	4
Cost efficiency	10%	5	3
Weighted score	100%	3.90 / 5	4.50 / 5

The weighted matrix selects Architecture B. The weights prioritize coverage, capacity, latency and reliability because the network must support critical applications as well as broadband. The macro-focused alternative would become more attractive if the dominant requirement were only low-cost wide-area coverage.

6.4 Network Slicing

Slice	Purpose	Priority	Resource policy
Slice 1 – eMBB	HD video and high-speed broadband	High	Large bandwidth, adaptive scheduling
Slice 2 – URLLC	Autonomous transport, industrial control, healthcare	Very high	Low-latency scheduling and reserved resources
Slice 3 – mMTC	Massive IoT	Medium	Efficient device access and energy-aware resources
Slice 4 – Public Safety	Emergency communications	Highest during incidents	Priority/pre-emption and resilient routing

Network slicing logically separates services while sharing common physical infrastructure. During congestion, critical slices receive protected resources so that public-safety and URLLC services are not degraded by ordinary broadband traffic.

6.5 Edge Computing Locations

Edge computing should be placed at major metropolitan areas, industrial clusters, transport corridors and critical healthcare/public-safety locations. The purpose is to move latency-sensitive processing closer to the user. Vehicle coordination, industrial control and remote healthcare applications can therefore avoid unnecessary round trips to a distant centralized cloud.

6.6 Failure and Demand Scenarios

Scenario	Effect	Response
Peak-hour congestion	High utilization and reduced throughput	AI traffic prediction, load balancing, small-cell activation and slice prioritization
Backhaul failure	Service degradation or isolation	Redundant links, alternate routing and local edge operation
Sudden emergency traffic	Public-safety demand increases sharply	Emergency slice priority, resource pre-emption and temporary capacity expansion
Edge failure	Latency-sensitive applications may degrade	Redundant edge nodes and controlled fallback to regional cloud

6.7 Design Decision 4

Architecture B is preferred because it provides the best balance for the consortium's mixed requirements. If the dominant requirement changes to rural coverage, the design would shift toward more sub-1 GHz macro deployment and fewer small cells. If the dominant requirement becomes ultra-reliable vehicle communication, the design would increase edge sites, redundant transport, dedicated URLLC resources and overlapping coverage along transport corridors.

7. Stage 5 – Integrated 5G/6G Network Design

7.1 Overall Architecture

The proposed end-to-end path is: User Equipment → Macro/Small-cell gNodeB → 5G Radio Access Network → Transport/Backhaul → 5G Core → UPF/Edge → Application/Cloud. AMF manages access and mobility

functions, SMF manages session-related control functions, and UPF handles user-plane traffic. Edge UPFs can provide local breakout for latency-sensitive services.

A simplified logical architecture is:

[UEs / Vehicles / IoT / Healthcare]
↓
[Sub-1 GHz Macro + Mid-band gNodeB + Small Cells + mmWave]
↓
[Transport / Redundant Backhaul]
↓
[5G Core: AMF | SMF | UPF | Policy / Authentication]
↓
[Regional Edge + Central Cloud]
↓
[Applications / Analytics / Emergency Services]

7.2 State-wise Radio Access Strategy

State	Macro layer	Small-cell layer	Edge strategy
Tamil Nadu	Sub-1 GHz + mid-band	Cities, factories, transport hubs	Industrial and urban edge
Karnataka	Sub-1 GHz + mid-band	Technology/enterprise districts	Urban and transport edge
Kerala	Sub-1 GHz emphasis + mid-band	Dense settlements and city centres	Healthcare/disaster-resilience edge
Maharashtra	Mid-band dense macro + sub-1 GHz	High-density urban and business zones	Multiple metropolitan edge sites
Gujarat	Sub-1 GHz + mid-band	Industrial corridors and ports	Industrial/logistics edge
Uttar Pradesh	Sub-1 GHz macro + mid-band cities	Selected urban hotspots	Regional edge near major cities/corridors

7.3 Mobility Management

- Maintain overlapping radio coverage on major highways and transport corridors.
- Configure neighbor-cell relationships and measurement thresholds carefully.
- Use beam tracking and rapid beam recovery for high-frequency layers.
- Use lower-frequency coverage as a fallback where mmWave blockage occurs.
- Place edge resources along high-mobility corridors so that sessions can remain close to users.
- Monitor handover success, interruption time and packet loss as mobility KPIs.

7.4 AI-Enabled Resource Management

The AI controller receives historical and real-time measurements such as traffic load, active users, cell utilization, radio quality and congestion. A prediction model forecasts near-future traffic. The controller then recommends spectrum allocation, user association, transmit-power adjustments, small-cell activation and traffic steering.

A practical control loop is: Measurement → Data preprocessing → Traffic prediction → Resource optimization → RAN configuration → KPI monitoring → Model update. A rule-based baseline can be retained so that the AI system can be evaluated against a known non-AI strategy.

7.5 Security and Resilience

Area	Proposed measure

Authentication	Strong subscriber/device authentication and access control
Encryption	Protect control and user traffic over applicable interfaces
Anomaly detection	AI-assisted detection of unusual traffic and signaling patterns
Network slicing security	Isolate slice resources and apply slice-specific policies
Backhaul resilience	Redundant paths and automatic failover
Core resilience	Redundant critical network functions and recovery procedures
Monitoring	Continuous KPI, fault and security-event monitoring
Recovery	Defined incident response, service restoration and fallback procedures

7.6 Sustainability and Energy Efficiency

- Use energy-saving sleep modes for lightly loaded small cells when service continuity permits.
- Use AI-based traffic prediction to avoid unnecessary always-on capacity.
- Share sites, towers, power systems and backhaul where technically and commercially feasible.
- Optimize transmit power and beam selection rather than operating all resources at maximum power.
- Use efficient cooling and power-management systems at edge and core facilities.
- Track energy per delivered bit as a deployment KPI.

7.7 Emerging 6G Concept

AI-native networking is selected as the primary emerging 6G concept. Future networks can use AI more deeply within radio, orchestration and service management. Integrated Sensing and Communication (ISAC) is another potential future capability for combining communication with sensing functions. Potential THz communications may provide extremely high data rates at very short ranges, but they require significant future technology development and are not assumed as a current deployment layer in this report.

The current design therefore distinguishes implementable 5G functions—5G NR, gNodeB, 5G Core, massive MIMO, beamforming, slicing and edge computing—from future 6G capabilities that require further standardization and validation.

8. Stage 6 – Validation and Final Deployment Decision

8.1 Validation Method

The design can be validated using MATLAB, Python or ns-3. The first model should calculate traffic demand, Erlang-B channel requirements, cell utilization and approximate cell counts. A second simulation can model user distribution, mobility, handover, propagation loss and interference. A third layer can compare AI-based resource allocation with a baseline scheduling strategy.

Validation output	**Method**	**Expected interpretation**
Traffic demand	Analytical calculation	Confirms busy-hour capacity requirement
Erlang B blocking	Erlang-B recursion	Checks traffic-channel requirement for target blocking
Cell utilization	Traffic/cell capacity	Identifies congestion
Coverage	Link-budget / propagation simulation	Checks geographical service availability
Throughput	System-level simulation	Measures eMBB performance

Latency	End-to-end budget/simulation	Checks URLLC and edge suitability
Handover	Mobility simulation	Checks transport continuity
Energy	Power model	Compares energy-saving strategies

8.2 Scenario 1 – Peak-Hour Congestion

During peak hours, eMBB traffic can increase significantly. The proposed response is to predict the load, activate available small-cell capacity, steer users between layers, allocate additional radio resources, and protect URLLC/public-safety slices. If congestion remains above the planning threshold, additional spectrum or physical densification is required.

8.3 Scenario 2 – Emergency / Infrastructure Failure

During an emergency, public-safety traffic receives priority. Redundant backhaul paths are used where available, unaffected edge sites continue local processing, and non-critical traffic can be deprioritized. The network should maintain a controlled fallback to the lower-frequency coverage layer if high-frequency links become unavailable.

8.4 AI Evaluation Against Baseline

The AI controller should be compared with a conventional fixed or rule-based allocation method. The evaluation should use the same traffic traces and constraints. Useful metrics are average throughput, 95th-percentile latency, packet loss, blocking probability, cell utilization balance and energy consumption. The AI approach is considered beneficial only if it improves one or more important KPIs without causing unacceptable degradation in other KPIs.

8.5 Specification–Verification Table

Requirement	**Design response**	**Evidence / validation**	**Status**
Wide-area coverage	Sub-1 GHz macro layer	Coverage/link-budget model	Proposed
Urban capacity	Mid-band + small cells	Cell utilization/throughput simulation	Proposed
High-capacity hotspots	Selective mmWave	Hotspot throughput model	Proposed
2% blocking target	Erlang-B channel calculation	Blocking probability calculation	Illustrative; replace inputs
Low latency	Edge computing + URLLC slice	Latency budget/simulation	Proposed
Mobility continuity	Overlap, handover and edge corridors	Mobility simulation	Proposed
Massive IoT	mMTC slice and efficient access	Device-load model	Proposed
Emergency resilience	Priority slice + redundancy	Failure scenario	Proposed
AI optimization	Prediction and resource controller	Baseline comparison	To validate
Energy efficiency	Sleep modes and adaptive resources	Energy model	To validate

8.6 Key Assumptions and Limitations

- The question paper specifies the six-state deployment and required technologies but does not provide numerical subscriber count, busy-hour traffic per subscriber or available traffic-channel values. The Erlang example therefore uses explicit illustrative assumptions.

- The cell capacity example is an assumed effective capacity and is not a measured operator value.
- Propagation loss, terrain, building penetration, antenna heights and regulatory spectrum assignments require site-specific data.
- Practical protocol, guard-band, duplexing and control overheads are not included in the idealized Erlang channel-count calculation.
- Actual 5G/6G deployment must comply with applicable spectrum licensing, power limits, safety requirements and infrastructure regulations.
- 6G concepts are treated as future capabilities and are not assumed to be fully deployable today.

9. Final Deployment Recommendation

Deployment with modifications is recommended. The proposed heterogeneous architecture is technically suitable for the consortium because it combines broad coverage, high capacity, low-latency processing, service isolation, mobility support and resilience. The design should be implemented in phases rather than as a uniform six-state rollout.

Phase	Deployment focus
Phase 1	Deploy sub-1 GHz and mid-band coverage in priority urban, rural and transport corridors; establish resilient 5G Core and backhaul.
Phase 2	Add small cells in congested cities, industrial clusters and transport hubs; deploy edge computing at critical locations.
Phase 3	Deploy mmWave selectively at justified high-capacity hotspots and enterprise/industrial sites.
Phase 4	Introduce AI-assisted optimization after sufficient network telemetry is collected and validated.
Phase 5	Evaluate emerging 6G concepts such as AI-native networking and ISAC through trials without assuming production readiness.

10. Conclusion

The six-state smart-city network requires a non-uniform cellular design because geography, population density, application demand and infrastructure conditions vary substantially. A layered heterogeneous architecture is therefore selected. Sub-1 GHz forms the coverage foundation, mid-band provides the principal capacity layer, and mmWave is used only where very high capacity justifies dense deployment.

OFDMA, massive MIMO, beamforming, frequency reuse and small-cell deployment improve capacity and spectrum utilization, while interference coordination is required to prevent the benefits of densification from being lost. Network slicing separates eMBB, URLLC, mMTC and public-safety requirements. Edge computing reduces latency for autonomous transportation, industrial automation and healthcare. AI-assisted resource management provides a path toward dynamic traffic prediction and optimization.

The quantitative examples demonstrate how traffic demand, spectral efficiency, cell capacity and Erlang-B blocking probability can be incorporated into engineering decisions. However, final deployment must replace illustrative assumptions with measured subscriber, traffic, spectrum, propagation and site data. With those modifications and validation through analytical or simulation tools, the proposed network provides a scalable pathway from current 5G capabilities toward future 6G-ready services.

11. Summary of Major Design Decisions

Decision	Recommended choice	Reason
Spectrum	Layered sub-1 GHz + mid-band + selective mmWave	Balances coverage, capacity and cost
RAN	Heterogeneous macro + small-cell architecture	Handles different traffic densities
Core	5G Core with AMF, SMF, UPF and edge integration	Supports scalable service delivery
Slicing	eMBB, URLLC, mMTC, public safety	Separates conflicting service requirements
Mobility	Overlapping coverage + optimized handover	Supports transport corridors
Interference	Beamforming, power control, coordination	Controls dense-network interference
AI	Traffic prediction and dynamic RRM	Improves adaptability
Security	Authentication, encryption, anomaly detection, redundancy	Protects critical services
Energy	Sleep modes, adaptive resources, infrastructure sharing	Reduces operational energy
Future 6G	AI-native networking / ISAC evaluation	Provides evolution path without assuming current readiness

12. Submission Note

This report follows the requested order of requirements → analysis → calculations → comparison → recommendation. Numerical values identified as illustrative should be replaced by actual planning inputs if they are provided by the instructor or operator.